

Shell reduction modulo 12, Chebyshev bias, and the structure of rough kernels

Abstract

We introduce a two-layer shell reduction of integers coprime to 6 and study the statistical behavior of the residual kernel under the action of the Dirichlet character χ_3 . Let $P_k = \{2, 3, 5, \dots, p_k\}$ and define for n with $\gcd(n, 6) = 1$ the quantities $S_k(n)$, $R_k(n)$, and the balance/switch points $\pi_{\text{bal}}^M(n)$, $\pi_{\text{wech}}^M(n)$.

Theorem A. For $n = \prod p_i^{a_i}$ and $M = R$, the points $\pi_{\text{bal}}^R(n)$ and $\pi_{\text{wech}}^R(n)$ are determined by the two critical indices of the cumulative sums of $a_i \log p_i$. In particular, for $n = pqr$ with $3 < p < q < r$ one has $\pi_{\text{bal}}^R = \pi_{\text{wech}}^R = q$.

Theorem B. If $k = k(N)$ satisfies $k \rightarrow \infty$ and $k = o(\log N)$, then the asymptotic densities of the four residue classes of $R_k(n) \bmod 12$ exist and satisfy

$$\omega_B > \omega_C > \omega_A > \omega_E,$$

with explicit rates $\omega_B - \omega_C \sim \delta / (\log p_k \log N)$ for $\delta > 0$ arising from the Chebyshev bias of χ_3 .

Theorem C. For prime quadruplets $Q_4 = (p, p+2, p+6, p+8)$ the functional $T_c = B_\pi + c \cdot \mathbf{1}_{EA-BC-EA}$ maximizes $\text{Corr}(T_c, \log \text{Gap}(Q_4))$ uniquely at $c = 6$, corresponding to the minimal strictly positive d with $\text{Cov}(\chi_3(p), \chi_3(p+d)) < 0$.

Theorem D. The expected waiting time for twin primes in a monochromatic EA or BC run of $R_k(n)$ equals $2^{4-k} - 1$, where k is the number of resolved shell primes.

The results unify numerical observations on prime constellations, Chebyshev bias, and entropy of residue classes into a single framework of logarithmic mass accounting.

1 Introduction

1.1 Motivation and definitions

Let $\mathcal{M} = \{n \in \mathbb{N} : \gcd(n, 6) = 1\}$. For $k \geq 1$ let $P_k = \{2, 3, 5, \dots, p_k\}$. For $n \in \mathcal{M}$ define

$$S_k(n) = \prod_{p \leq p_k} p^{v_p(n)}, \quad R_k(n) = \frac{n}{S_k(n)}.$$

We factor $R_k(n)$ according to residues mod 12:

$$\begin{aligned} E_k(n) &= \prod_{\substack{q^a \parallel R_k \\ q \equiv 1 \pmod{12}}} q^a, & A_k(n) &= \prod_{\substack{q^a \parallel R_k \\ q \equiv 5 \pmod{12}}} q^a, \\ B_k(n) &= \prod_{\substack{q^a \parallel R_k \\ q \equiv 7 \pmod{12}}} q^a, & C_k(n) &= \prod_{\substack{q^a \parallel R_k \\ q \equiv 11 \pmod{12}}} q^a. \end{aligned}$$

For $M \in \{E, A, B, C, EA, BC, ABC, R\}$ set M_k the corresponding product, with $R_k = E_k A_k B_k C_k$.

Definition 1.1. For $n \in \mathcal{M}$ and a mode M define $L_k^M(n) = \log M_k(n) - \log S_k(n)$. Then

$$\begin{aligned} \pi_{\text{bal}}^M(n) &:= p_j, \quad j = \arg \min\{k : S_k > 1, M_k > 1, |L_k^M|\}, \\ \pi_{\text{wech}}^M(n) &:= p_j, \quad j = \min\{k : L_{k-1}^M > 0, L_k^M \leq 0, L_k^M \neq 0\}. \end{aligned}$$

For $X \in \{E, A, B, C\}$ and x, y set

$$\psi_X(x, y) := \sum_{\substack{n \leq x \\ p_{\min}(n) > y \\ P^+(n) \equiv x_X \pmod{12}}} \log P^+(n),$$

and $W_X^{(k)}(N)$ the normalized mean of $\log X_k(n)$ over $n \leq N, n \in \mathcal{M}$.

2 Shell arithmetic and the balance/switch theorems

Lemma 2.1 (Non-triviality). $S_k(n) > 1$ iff $p_{\min}(n) \leq p_k$. $M_k(n) > 1$ iff $R_k(n)$ has a prime factor in the residue classes of M .

Theorem 2.2 (Theorem A, general case). Let $n = \prod_{i=1}^t p_i^{a_i}$ with $3 < p_1 < \dots < p_t$ and $A_i := a_i \log p_i$. Let $M = R$. Then:

1. Let j be minimal with $A_j \geq \sum_{i>j} A_i$. Then $\pi_{\text{bal}}^R(n) = p_j$.
2. Let ℓ be minimal with $\sum_{i \leq \ell} A_i \geq \sum_{i>\ell} A_i$. Then $\pi_{\text{wech}}^R(n) = p_\ell$.

Proof. For $k_{i-1} \leq m < k_i$ we have $L_m^R = \sum_{j \geq i} A_j - \sum_{j < i} A_j$. The minimal $|L_m^R|$ under non-triviality occurs at the first i with $A_i \geq \sum_{j>i} A_j$. The first sign change occurs when the partial sum crosses the remaining sum. $\square$

Corollary 2.3. For $n = pqr$ with $3 < p < q < r$ one has $\pi_{\text{bal}}^R = \pi_{\text{wech}}^R = q$.

3 Asymptotic hierarchy of residue classes

Theorem 3.1 (Theorem B). Let $k = k(N)$ with $k \rightarrow \infty$, $k = o(\log N)$ and $y = p_k$. Then limits $\omega_X = \lim_{N \rightarrow \infty} W_X^{(k)}(N)$ exist and satisfy

$$\omega_B > \omega_C > \omega_A > \omega_E.$$

Moreover,

$$\omega_B - \omega_C \sim \frac{\delta}{\log y \log N},$$

where $\delta > 0$ depends on the Chebyshev bias for χ_3 .

Sketch. For $1 - o(1)$ of $n \in \mathcal{M}_N$, $\log R_k(n) = \log P^+(n) + O(\log y)$. Hence

$$W_B - W_C \sim \frac{\psi_B(N, y) - \psi_C(N, y)}{\sum_X \psi_X(N, y)}.$$

Using Buchstab and $\sum_{p \leq t, p \equiv 7} \log p - \sum_{p \leq t, p \equiv 11} \log p \sim \delta_0 \sqrt{t}$, partial summation gives the claim. $\square$

4 Applications to prime k -tuples

Theorem 4.1 (Theorem C). For prime quadruplets $Q_4 = (p, p+2, p+6, p+8)$ let $T_c = B_\pi + c \cdot \mathbf{1}_{EA-BC-EA}$. Then

$$\arg \max_{c \in \mathbb{N}} \lim_{x \rightarrow \infty} \text{Corr}(T_c, \log \text{Gap}(Q_4)) = 6,$$

because $d = 6$ is minimal with $\text{Cov}(\chi_3(p), \chi_3(p+d)) < 0$.

Sketch. B_π is χ_3 -neutral by Theorem A. The indicator encodes $\text{sgn} \sum_{n \in Q_4} \chi_3(n) \log n$. For $h = (0, 2, 6, 8)$ the dominant covariance term is from distance 6, which is maximally negative. Optimizing the linear predictor gives $c^* = 6$. $\square$

5 Entropy and waiting times

Theorem 5.1 (Theorem D). Let $Z_n = \mathbf{1}_{BC}(R_k(n))$. Then (Z_n) is stationary with $\text{Cov}(Z_n, Z_{n+h}) = \frac{1}{4} \rho^h + o(1)$, $\rho = 1 - 2^{-k}$. The expected waiting time for a twin in EA or BC equals $2^{4-k} - 1$.

Sketch. Conditioning on the first k primes, $\mathbb{P}[Z_{n+2} = 0 \mid Z_n = 0] = \frac{1}{2} + 2^{-k-1}$. The mean recurrence time for 00 with gap 2 gives $\mathbb{E} = 2^{4-k} - 1$. $\square$

6 Concluding remarks

Theorems A–D reduce the observed $B > C > A > E$ hierarchy, the $c^* = 6$ trigger, and the $2^{4-k} - 1$ law to logarithmic mass accounting, Chebyshev bias, and entropy. Further work concerns unconditional error terms and higher moduli.

References

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